

Experimental Psychology 30 - Assignment 2

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Part One: Multiple-Choice Questions (8 marks)

Letter	Statement
All Case Studies used in this assignment are summarized from <i>Mindless Eating</i> by Brian Wansink. Use <i>Case Study A</i> to answer multiple-choice questions 1, 2, and 3. Case Study A Billy, a galley cook on a Navy ship, was facing daily complaints from the crew about the lemon Jell-O he was serving. Because of an ordering error, lemon was the only flavour of Jell-O purchased for consumption during the four-month exercise. There was no cherry Jell-O and the crew was unhappy. Because the ship was not going to be in port (land) for another two months, Billy had to act in a creative manner – he added red food colouring to the lemon Jell-O. When the red-coloured lemon Jell-O was served to the sailors, no one complained.	
B	1. The manipulated variable in this Case Study is A. sailors at sea B. Jell-O colour C. time away from port D. satisfaction level of the sailors
D	2. The responding variable in this Case Study is A. sailors at sea B. Jell-O colour C. time away from port D. satisfaction level of the sailors

ANSWER KEY

A	3. The sailors believed the red Jell-O was cherry-flavoured Jell-O, not lemon-flavoured Jell-O. Psychologists refer to this phenomenon as A. expectation assimilation B. cognitive assumption C. brainwashing D. bias anticipation
Use <i>Case Study B</i> (below) to answer multiple-choice questions 4 and 5. <i>Case Study B</i> One hundred forty (140) male college students were asked to read a summary of a fictitious date between Brad and Barb. The students were divided into two groups of 70 people each. The summary described going to a movie and gave much detail about what Brad did on the date. The only difference in the descriptions that each group was given related to the amount of popcorn Brad ate during the movie. In one version of the summary Brad ate "a couple of handfuls" of his popcorn. In the other version Brad ate "almost all" of his popcorn. Researchers asked the two groups of men how much weight (in pounds) they thought Brad could bench press. Students who read that Brad ate almost all of his popcorn imagined Brad lifting, on average, 21 more pounds than Brad who only ate a couple of handfuls of his popcorn.	
A	5. The manipulated variable in this Case Study was the A. amount of popcorn Brad ate B. amount of weight Brad was thought to bench press C. type of movie Brad was watching while eating the popcorn D. level of hunger Brad was experiencing during the movie
B	6. The responding variable in this Case Study is A. amount of popcorn Brad ate B. amount of weight Brad was thought to bench press C. type of movie Brad was watching while eating the popcorn D. level of hunger Brad was experiencing during the movie

ANSWER KEY

A	6. If you expect a food to taste bad, the food will taste bad. This phenomenon is known by two names, one of which is A. confirmation bias B. foregone conclusion C. cognitive anticipation D. assimilation integration
B	7. A good hypothesis contains both the A. controlled and manipulated variables B. responding and manipulated variables C. manipulated variable and control group D. responding variable and materials
<i>Use the following information to answer multiple-choice question 8 and short-answer question 1.</i> <i>Case Study B - Extended</i> The researchers from <i>Case Study B</i> gave the same two versions of the "Brad and Barb date summary" to two groups of female college students. The women were also asked how much weight (in pounds) they thought Brad could bench press. Unlike their male counterparts, however, the women noted no significant difference in the amount of weight Brad might lift.	
C	8. A researcher reviewed the differences between the male and female responses in Case Study B. In this comparison of data, the manipulated variable was the _____ and the responding variable was the _____. A. amount of popcorn Brad ate; gender of the college students B. amount of weight Brad was thought to bench press; gender of the college students C. gender of the college students; amount of weight Brad was thought to bench press D. gender of the college students; amount of popcorn Brad ate

Part Two: Written-Response Questions (13 marks)

ANSWER KEY

1. In Case Study B – Extended, why do you think the responses of the men and the responses of the women were different? **(2 marks)**

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2. When deciding on a problem for an experiment, researchers conduct background research. Give two reasons background research is important. **(2 marks)**

- to learn more about the subject area
- to ensure the question has not already been answered by someone else
- may change or refine the focus of an experiment
- learn about the techniques and equipment others have used in similar experiments
- to learn from the experience of others and avoid their mistakes.

3. Describe the importance of the control group in an experiment. **(2 marks)**

Good experiments compare different groups of trials with each other. This comparison is done using a Control group to ensure that the changes observed are caused by the independent variable.

4. Larry performs an experiment on his two best friends. Explain why his sample size is too small. **(2 marks)**

More than one test subject is used for much the same reason more than one trial is performed. Scientists want results that are not considered flukes or anomalies. The sample size must be large enough to rule out such extremes or abnormalities in individuals.

5. Define sensory specific satiety and provide an example other than those given in the course content. **(3 marks)**

sensory specific satiety: a condition where one of five senses becomes overstimulated and no longer registers the stimulus as new - should not be examples involving Jelly Beans, M&Ms or buffet food.

ANSWER KEY

6. In Case Study Six (Rainforest Smoothies), how did the mother get her son to eat his broccoli? (2 marks)

She had convinced them that broccoli looked like a dinosaur tree and when they ate broccoli, they could pretend they were “long-necked dinosaurs eating the dinosaur trees.” At the dinosaur loving age of three and five, that was pretty cool, and it quickly became pretty cool to their friends.

Part Three: Scientific Method – Procedure (10 marks)

1. Section Four of the course described an experiment using yogurt. Researchers wanted to learn if the expectations of participants affected their actual taste experience. Below is a summary of the pertinent information.

A group of 32 individuals were invited to sample some new strawberry yogurt. The test subjects were seated in a food lab and the lights were turned off. They were then given chocolate yogurt. Of the 32 people, 19 rated the chocolate yogurt as having a distinctive strawberry taste.

Using the information, write a detailed procedure so that another person would be able to replicate the experiment. (10 marks)

IMPORTANT: You only need to write a procedure for the experiment!

----- End of Assignment Two -----

When you have completed your assignment, upload this document to the Assignment 2 dropbox on Brightspace.